SOMAIYA VIDYAVIHAR

K J Somaiya Institute of Technology

An Autonomous Institute Permanently Affiliated to the University of Mumbai

**DEPARTMENT OF INFORMATION TECHNOLOGY**

**Name:** Vaishnavi More

**Roll Number:** B - 72

**Batch:** SB4

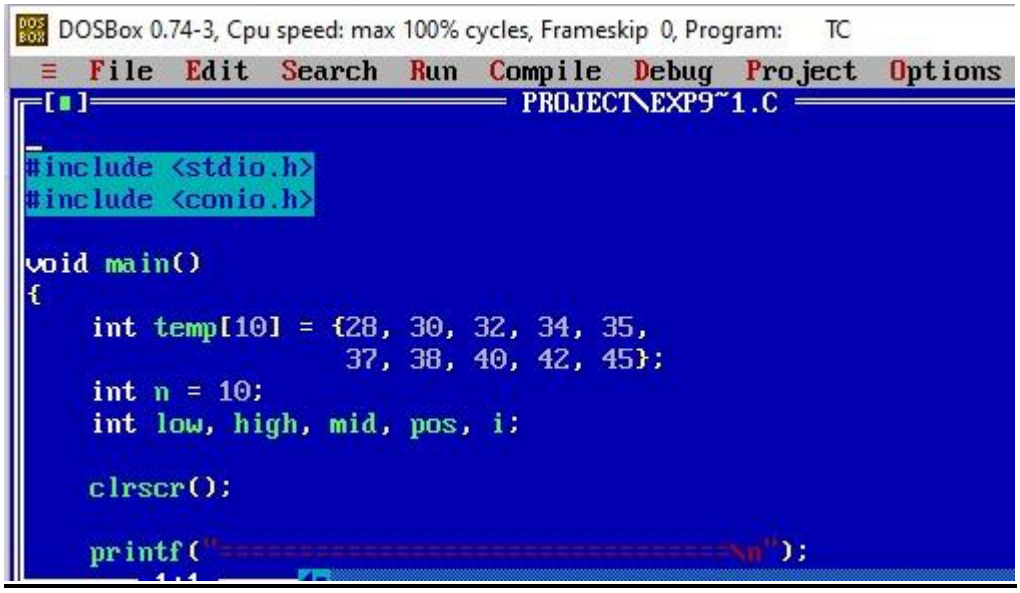
**Course Name and Code**: Data Structures Lab (ITL302)

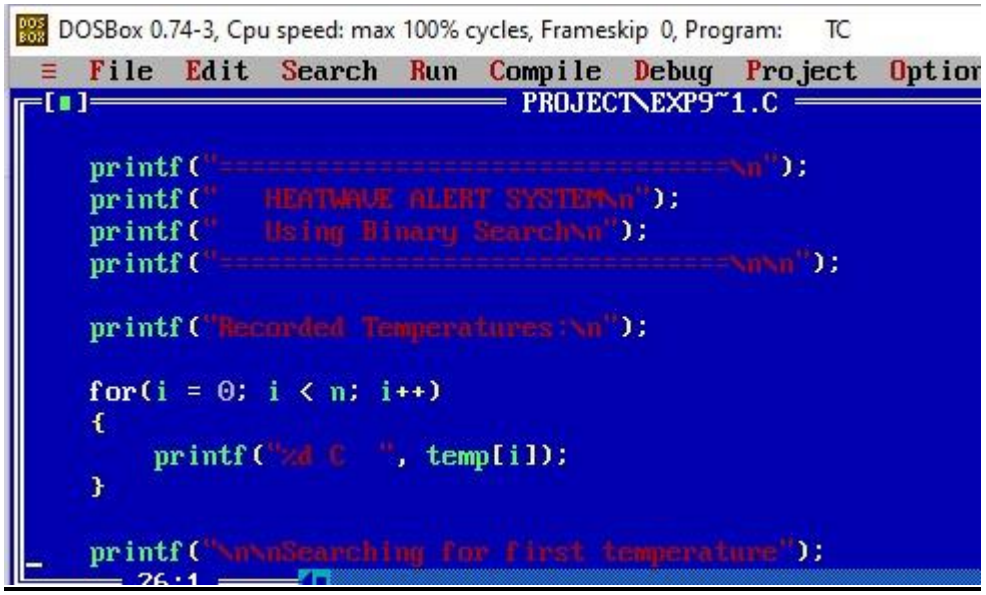
**Semester**: III (SYIT)

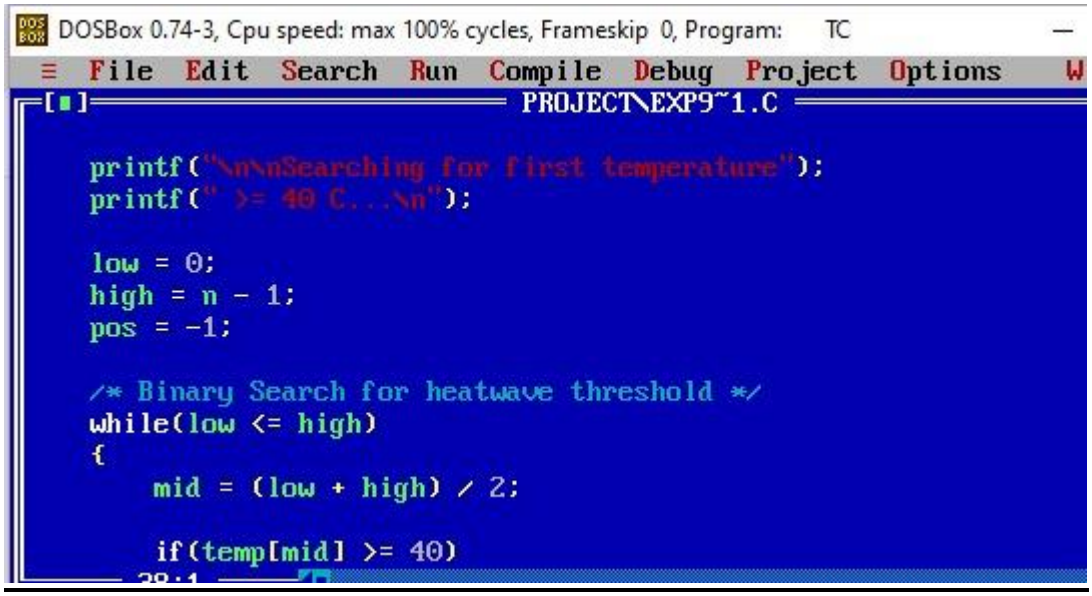
**Academic Year**: 2026-27 (Odd Semester)

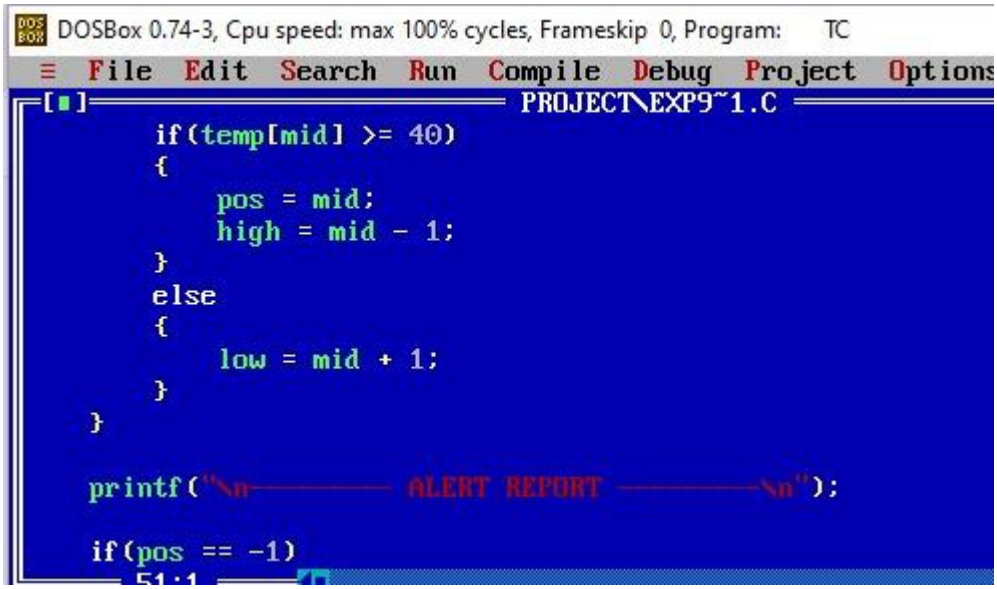
## EXPERIMENT 9

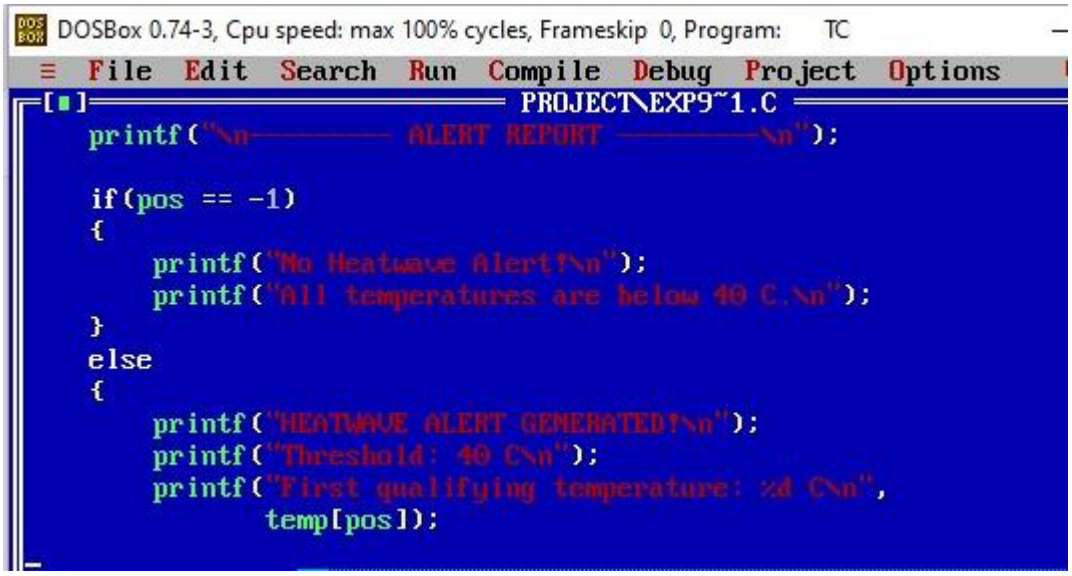
**Code**

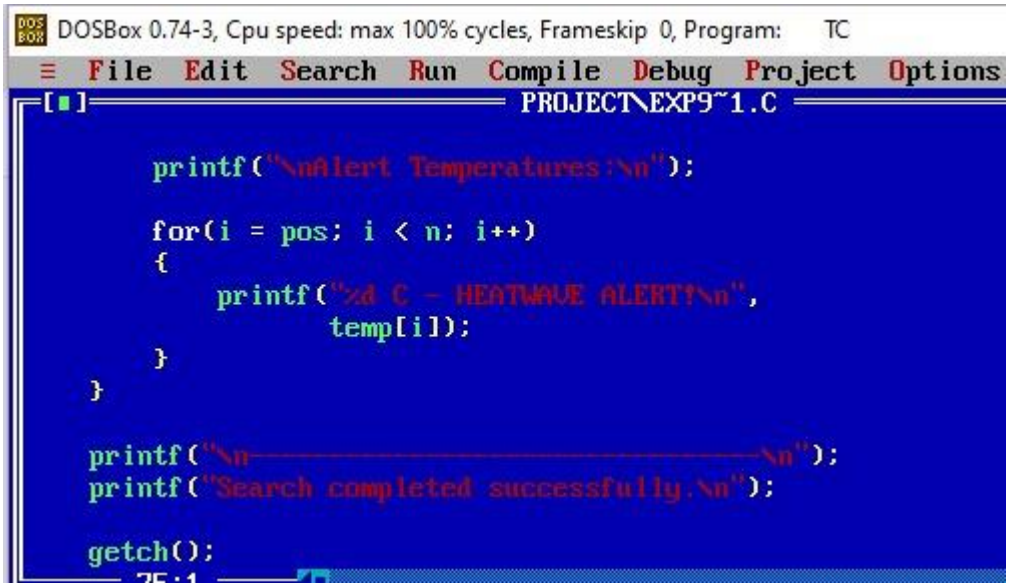


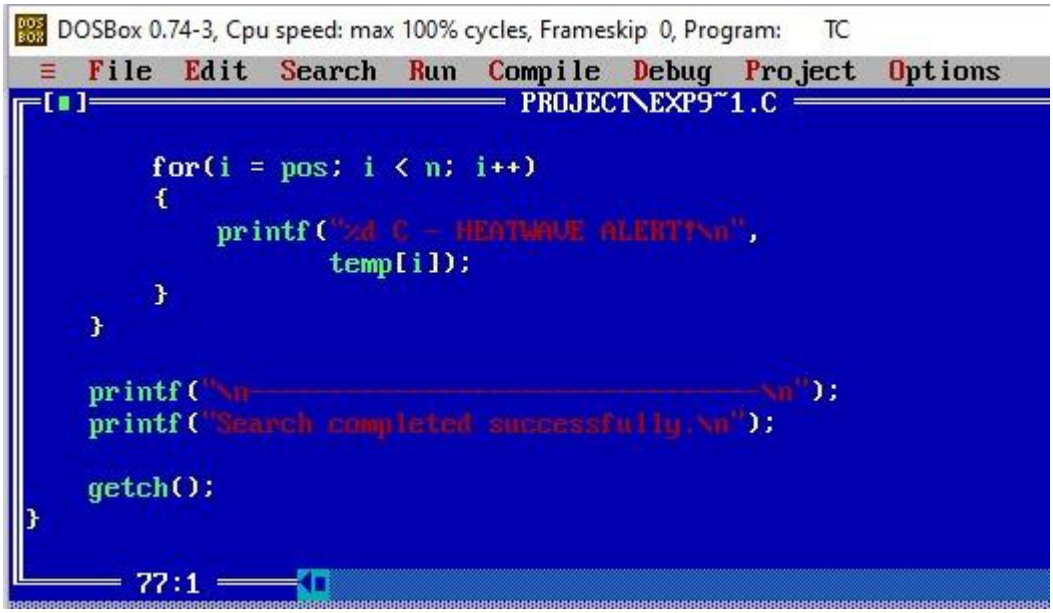

DOSBox 0.74-3, Cpu speed: max 100% cycles, Frameskip 0, Program: TC
File Edit Search Run Compile Debug Project Optior
PROJECT\EXP9~1.C
printf("==========\n");
printf(" HEATWAVE ALERT SYSTEM\n");
printf(" Using Binary Search\n");
printf("==========\n\n");
printf("Recorded Temperatures:\n");
for(i = 0; i < n; i++)
{
printf("%d C ", temp[i]);
}
printf("\n\nSearching for first temperature");


DOSBox 0.74-3, Cpu speed: max 100% cycles, Frameskip 0, Program: TC
File Edit Search Run Compile Debug Project Options W
PROJECT\EXP9~1.C
printf("\n\nSearching for first temperature");
printf(" >= 40 C...\n");
low = 0;
high = n - 1;
pos = -1;
/* Binary Search for heatwave threshold */
while(low <= high)
{
mid = (low + high) / 2;
if(temp[mid] >= 40)

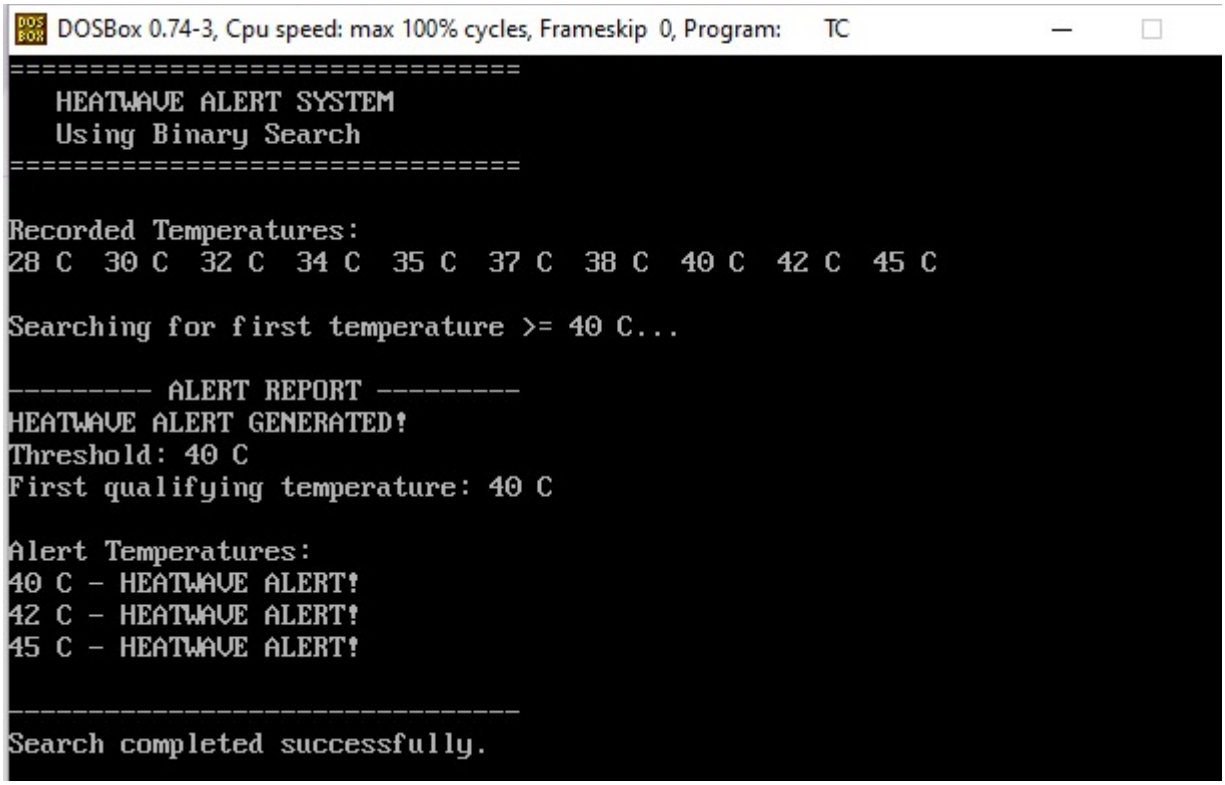

DOSBox 0.74-3, Cpu speed: max 100% cycles, Frameskip 0, Program: TC
File Edit Search Run Compile Debug Project Options
PROJECT\EXP9~1.C
if(temp[mid] >= 40)
{
pos = mid;
high = mid - 1;
}
else
{
low = mid + 1;
}
}
printf("\n---------- ALERT REPORT ----------\n");
if(pos == -1)


DOSBox 0.74-3, Cpu speed: max 100% cycles, Frameskip 0, Program: TC
File Edit Search Run Compile Debug Project Options
PROJECT\EXP9~1.C
printf("\n---------- ALERT REPORT ----------\n");
if(pos == -1)
{
printf("No Heatwave Alert!\n");
printf("All temperatures are below 40 C.\n");
}
else
{
printf("HEATWAVE ALERT GENERATED!\n");
printf("Threshold: 40 C\n");
printf("First qualifying temperature: %d C\n",
temp[pos]);


DOSBox 0.74-3, Cpu speed: max 100% cycles, Frameskip 0, Program: TC
File Edit Search Run Compile Debug Project Options
PROJECT\EXP9~1.C
printf("\nAlert Temperatures:\n");
for(i = pos; i < n; i++)
{
printf("%d C - HEATWAVE ALERT!\n",
temp[i]);
}
}
printf("\n----------\n");
printf("Search completed successfully.\n");
getch();


DOSBox 0.74-3, Cpu speed: max 100% cycles, Frameskip 0, Program: TC
File Edit Search Run Compile Debug Project Options
PROJECT\EXP9~1.C
for(i = pos; i < n; i++)
{
printf("%d C - HEATWAVE ALERT!\n",
temp[i]);
}
}
printf("\n----------\n");
printf("Search completed successfully.\n");
getch();
}
77:1

**output**


DOSBox 0.74-3, Cpu speed: max 100% cycles, Frameskip 0, Program: TC
==========
HEATWAVE ALERT SYSTEM
Using Binary Search
==========
Recorded Temperatures:
28 C 30 C 32 C 34 C 35 C 37 C 38 C 40 C 42 C 45 C
Searching for first temperature >= 40 C...
--------- ALERT REPORT ---------
HEATWAVE ALERT GENERATED!
Threshold: 40 C
First qualifying temperature: 40 C
Alert Temperatures:
40 C - HEATWAVE ALERT!
42 C - HEATWAVE ALERT!
45 C - HEATWAVE ALERT!
----------
Search completed successfully.